



Assignment

Spring 2026 2nd Semester

Course Title: Circuit Analysis Lab

Course Code: CMPE-121L

Topic(s): Lab 3 Introduction to Multimeter and Basic Series Circuit Implementation

SUBMITTED TO:

Mr. Mehmood-ul-Haq

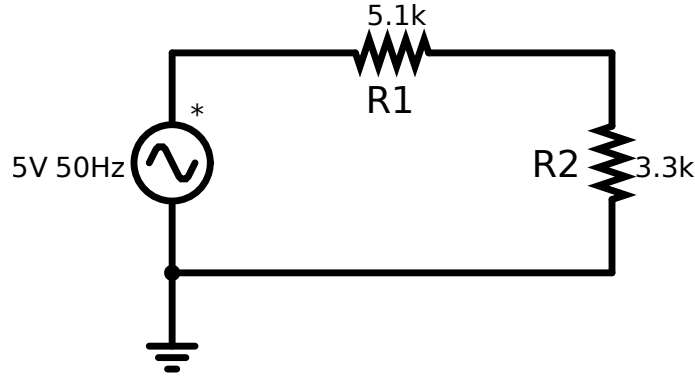
SUBMITTED BY:

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Submission Date: 31 Mar, 2026

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Main Campus

Experiment 1



Practical Procedure

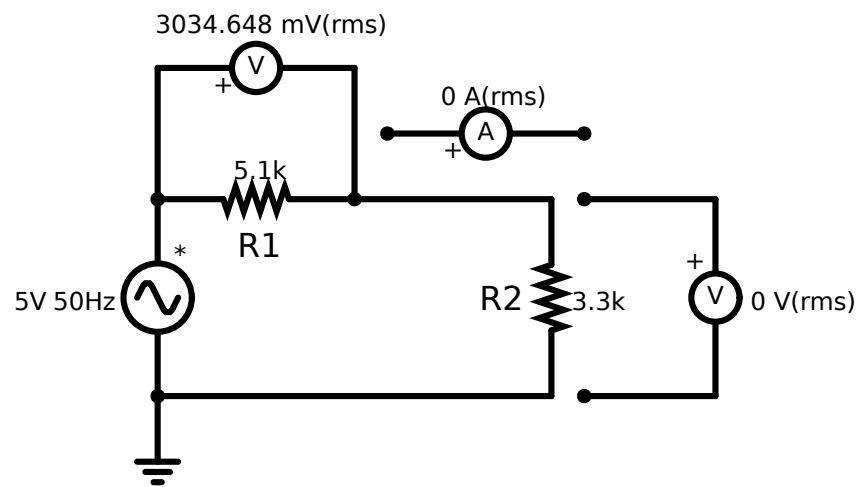
The objective of this experiment is to analyze the behavior of an AC series circuit consisting of two resistors: $R_1 = 5.1 \text{ k}\Omega$ and $R_2 = 3.3 \text{ k}\Omega$, driven by a 5 V alternating current source at a frequency of 50 Hz. For measurements, we utilize the **UNI-T UT89XE** digital multimeter to capture the RMS voltage drops and the total circuit current. Due to the virtual nature of the current lab sessions, the circuit is constructed and analyzed using the **Falstad CircuitJS** to simulate the circuit and measurements.

To measure the RMS voltages V_{R1} and V_{R2} , the multimeter dial is set to the **2V** or **20V AC** range, depending on the specific magnitude of the measurement taken (AC indicated by the **V~** symbol). The black probe remains in the **COM** port while the red probe is inserted into the **V Ω Hz** terminal. The probes are placed in **parallel** across each resistor to capture the potential difference. For the current measurement, the red probe is relocated to the **mA μ A** terminal, and the dial is set to the **2000 μ A AC** range (indicated by the **A~** symbol). Measuring current requires the DMM to be connected in **series**, effectively becoming part of the conductive path to allow the alternating charge flow to pass through the meter's internal shunt.

Theoretical calculations for AC circuits assume ideal waveforms and perfect components; however, simulated and physical results may vary. Factors such as the **internal impedance** of the multimeter and the **loading effect** of the voltmeter (typically $10 \text{ M}\Omega$) can slightly alter the circuit's total impedance. Furthermore, the precision of the **50Hz** frequency and the inherent **tolerance** of the resistors (R_1 and R_2) play a significant role in the minor discrepancies observed between the mathematical RMS predictions and the recorded experimental data.



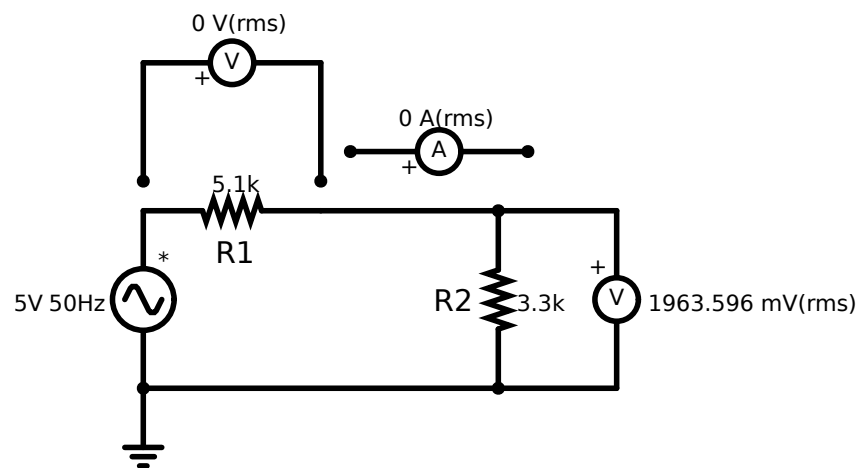
DMM: Dial to **20V AC**



Falstad: Voltage across R_1



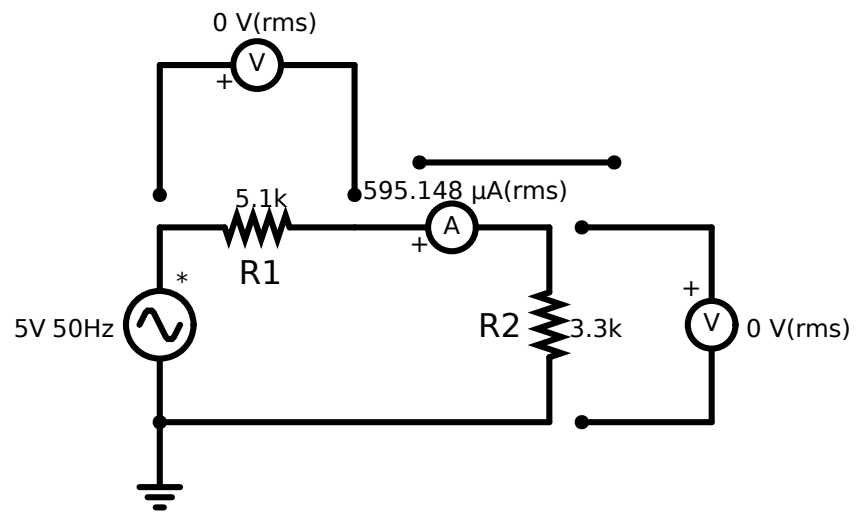
DMM: Dial to **2V AC**



Falstad: Voltage across R_2



DMM: Dial to
2000 μ A AC



Falstad: Current between R_1 and R_2

Theoretical Calculations

The following calculations assume an ideal AC source with $V_{in(RMS)} = 5\text{ V}$ and $f = 50\text{ Hz}$.

Total Impedance, Z_{total}

For the series circuit containing $R_1 = 5.1\text{ k}\Omega$ and $R_2 = 3.3\text{ k}\Omega$:

$$Z_{total} = R_1 + R_2 = 5.1\text{ k}\Omega + 3.3\text{ k}\Omega = 8.4\text{ k}\Omega$$

RMS Voltage across R_1 , V_{R1}

$$V_{R1(RMS)} = V_{in(RMS)} \cdot \left(\frac{R_1}{Z_{total}} \right) = 5\text{ V} \cdot \left(\frac{5.1\text{ k}\Omega}{8.4\text{ k}\Omega} \right) \approx 3.03571\text{ V}$$

RMS Voltage across R_2 , V_{R2}

$$V_{R2(RMS)} = V_{in(RMS)} \cdot \left(\frac{R_2}{Z_{total}} \right) = 5\text{ V} \cdot \left(\frac{3.3\text{ k}\Omega}{8.4\text{ k}\Omega} \right) \approx 1.96429\text{ V}$$

Total RMS Current , I_{RMS}

$$I_{RMS} = \frac{V_{in(RMS)}}{Z_{total}} = \frac{5\text{ V}}{8400} \approx 0.59524\text{ mA}$$

Experiment 1 Data

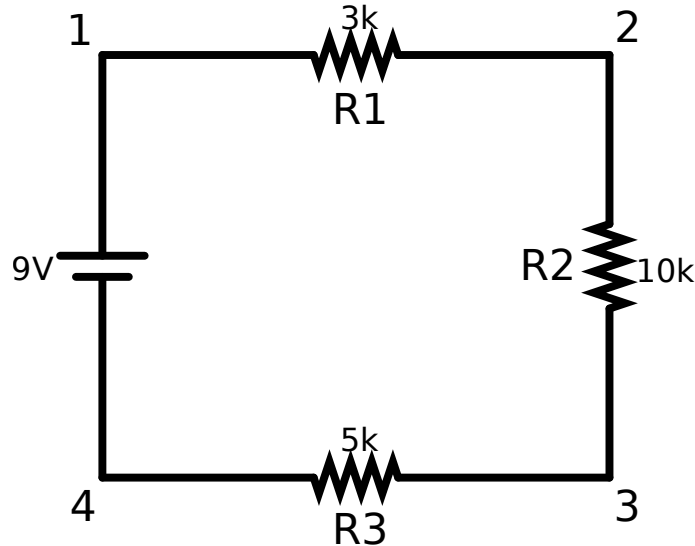
Practical Values (Simulated)

R1 Voltage	R2 Voltage	Current
3.03465 V	1.96360 V	0.59515 mA

Theoretical Values

R1 Voltage	R2 Voltage	Current
3.03571 V	1.96429 V	0.59524 mA

Experiment 2



Practical Procedure

The objective of this experiment is to analyze the behavior of a DC series circuit consisting of three resistors: $R_1 = 3\text{ k}\Omega$, $R_2 = 10\text{ k}\Omega$, and $R_3 = 5\text{ k}\Omega$, driven by a constant **9V DC** source. For physical measurements, we will be using the **UNI-T UT89XE** digital multimeter to capture the potential differences and branch currents at specific junctions, namely **Node 2** (positioned between R_1 and R_2) and **Node 3** (positioned between R_2 and R_3). Due to the virtual nature of the current lab sessions, the circuit is constructed and analyzed using the **Falstad CircuitJS** to simulate the circuit and measurements.

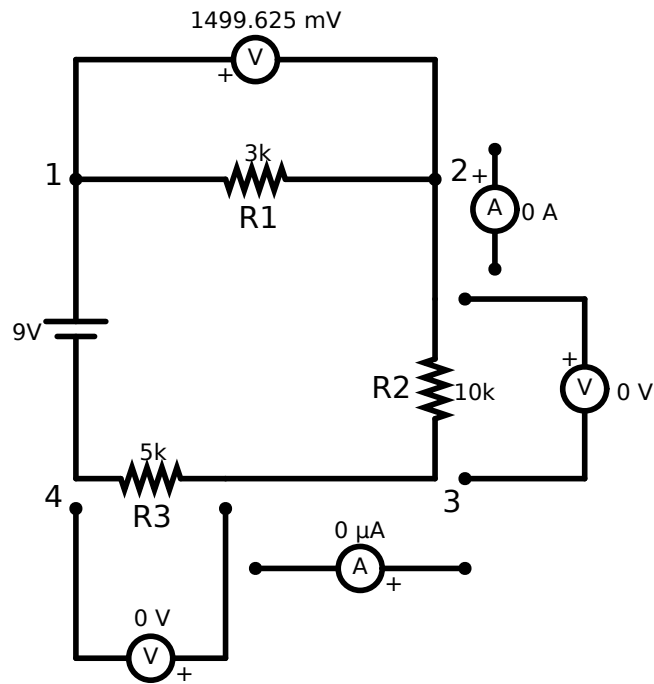
To capture the voltages V_{R1} , V_{R2} , and V_{R3} , the multimeter is set to the **2V** or **20V DC** range, depending on the specific magnitude of the measurement taken (DC indicated by the **V—** symbol). The black probe is inserted into the **COM** port, and the red probe is placed in the **VΩHz** terminal. Measurements are taken by placing the probes in **parallel** across each individual resistor. For current measurements (I_{Node2} and I_{Node3}), the red probe must be relocated to the **mA μA** terminal and the dial set to the **2000μA DC** range (indicated by the **A—** symbol). Unlike voltage, current is measured in **series**; this requires physically breaking the circuit at the node and inserting the DMM into the loop so that the charge carriers are forced to flow through the meter's internal sensing circuitry.

It is important to note that while theoretical calculations assume ideal components, practical readings often exhibit slight deviations. This is primarily due to

the **internal resistance** of the measuring instrument; the voltmeter has a finite high resistance that may slightly load the circuit, while the ammeter possesses a non-zero burden voltage. Additionally, the **tolerance** of the physical resistors (typically $\pm 5\%$) and the precise calibration of the **9V** supply contribute to the variance between mathematical predictions and the recorded data.



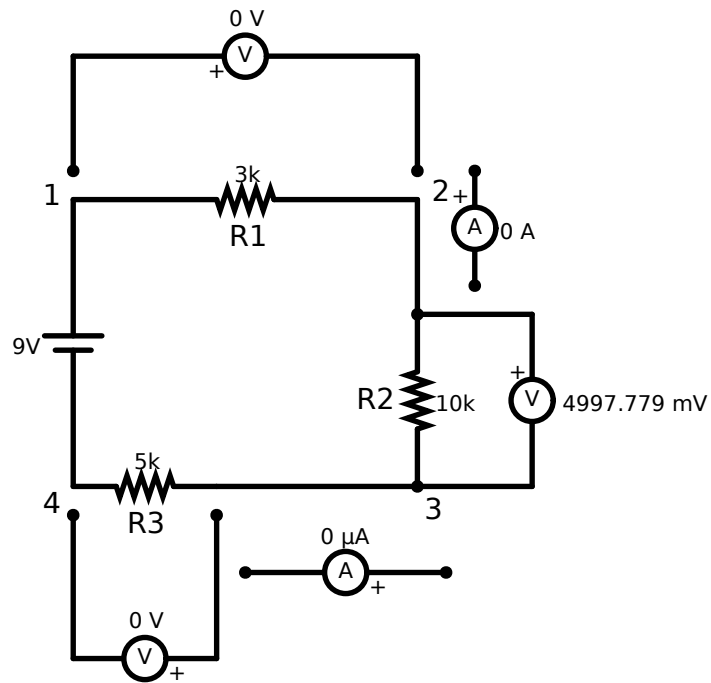
DMM: Dial to **2V DC**



Falstad: Voltage across R_1



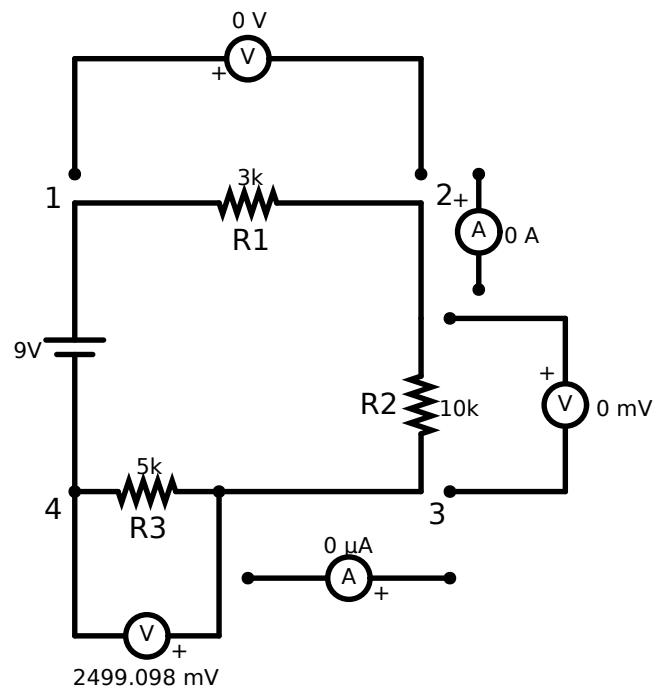
DMM: Dial
to 20V DC



Falstad: Voltage across R_2



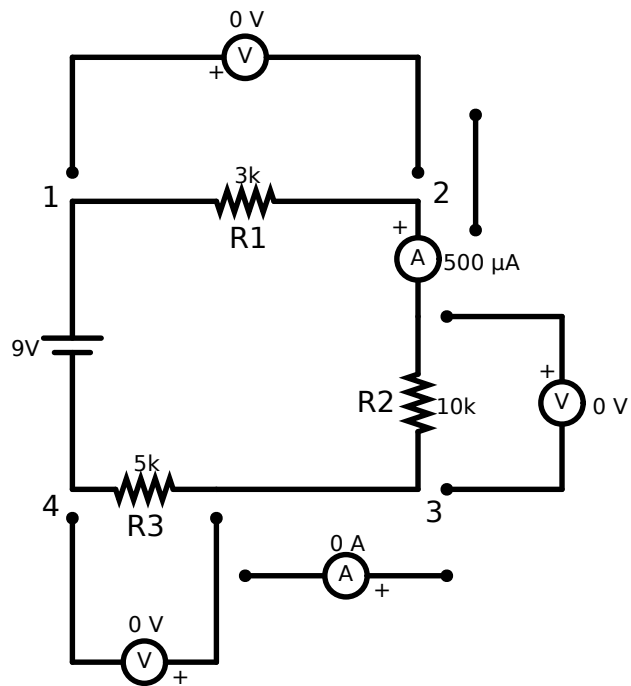
DMM: Dial
to 20V DC



Falstad: Voltage across R_3



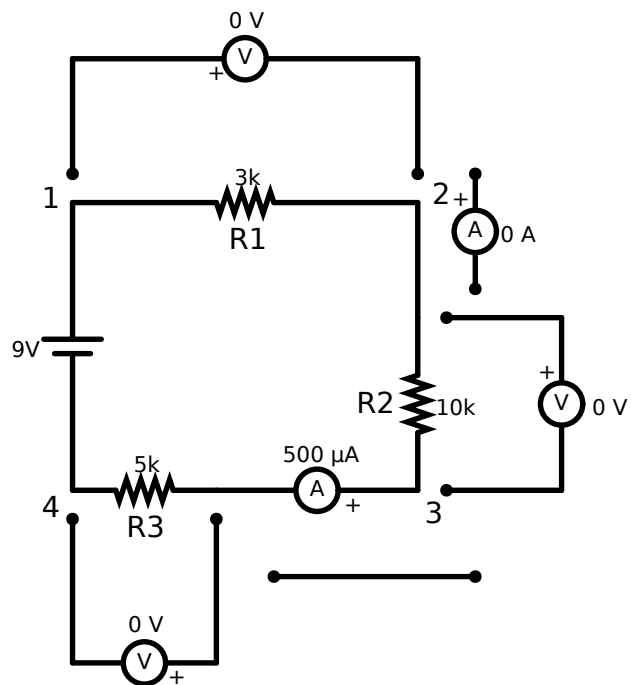
DMM: Dial to
2000μA DC



Falstad: Current through Node 2



DMM: Dial to
2000μA DC



Falstad: Current through Node 3

Theoretical Calculations

The circuit consists of three resistors in series: $R_1 = 3\text{ k}\Omega$, $R_2 = 10\text{ k}\Omega$, and $R_3 = 5\text{ k}\Omega$, powered by a $V_{in} = 9\text{ V}$ DC source.

Total Resistance, R_{total}

$$R_{total} = R_1 + R_2 + R_3 = 3\text{ k}\Omega + 10\text{ k}\Omega + 5\text{ k}\Omega = 18\text{ k}\Omega$$

Total Circuit Current, I_{total}

$$I_{total} = \frac{V_{in}}{R_{total}} = \frac{9\text{ V}}{18\text{ k}\Omega} = 0.5\text{ mA}$$

Current I_{Node2} at node 2 and I_{Node3} at node 3

In a series circuit, the current is common throughout the loop ($I_{total} = I_{R1} = I_{R2} = I_{R3}$). Therefore:

$$I_{Node2} = I_{Node3} = I_{total} = 0.5\text{ mA}$$

Individual Resistor Voltages

We apply Ohm's Law ($V = I \cdot R$).

$$V_{R1} = 0.5\text{ mA} \cdot 3\text{ k}\Omega = 1.5\text{ V}$$

$$V_{R2} = 0.5\text{ mA} \cdot 10\text{ k}\Omega = 5.0\text{ V}$$

$$V_{R3} = 0.5\text{ mA} \cdot 5\text{ k}\Omega = 2.5\text{ V}$$

Kirchhoff's Voltage Law (KVL) Validation

To ensure accuracy, we verify that the sum of the voltage drops equals the source voltage:

$$\sum V = 1.5\text{ V} + 5.0\text{ V} + 2.5\text{ V} = 9.0\text{ V}$$

Experiment 2 Data

Practical Values (Simulated)

R_1 Voltage	R_2 Voltage	R_3 Voltage	Current at node 2	Current at node 3
1.49963 V	4.99778 V	2.49910 V	0.50000 mA	0.50000 mA

Theoretical Values

R_1 Voltage	R_2 Voltage	R_3 Voltage	Current at node 2	Current at node 3
1.50000 V	5.00000 V	2.50000 V	0.50000 mA	0.50000 mA

Post-Lab Questions

1. Define energy.

Energy is the capacity or ability to do work, measured in Joules (J). In an electrical context, it represents the total electrical charge moved through a potential difference over a specific period of time.

2. Define power.

Power is the rate at which energy is consumed or transformed within a circuit. It is measured in Watts (W) and is calculated as the product of voltage and current ($P = V \cdot I$).

3. What is charge?

Charge is a fundamental physical property of matter that causes it to experience a force when placed in an electromagnetic field. In electrical circuits, the flow of these charges (typically electrons) constitutes an electric current.

4. What is a circuit?

A circuit is a closed-loop path through which an electric current can flow. It typically consists of a power source (like a battery), conducting wires, and loads (like resistors) that utilize the electrical energy.

5. What is the role of a resistor in the series circuit?

The primary role of a resistor is to oppose or limit the flow of electric current. In a series circuit, resistors also act as voltage dividers, where the total source voltage is distributed across each resistor based on its ohmic value.

6. What is the voltage divider rule?

The Voltage Divider Rule (VDR) is a formula used to calculate the voltage drop across a specific resistor in a series circuit without needing to calculate the total current first. It states that the voltage across a resistor (R_n) is the total voltage multiplied by the ratio of that resistance to the total resistance:

$$V_{Rn} = V_{total} \cdot \frac{R_n}{R_{total}}$$

7. What are the features of a series circuit?

The key features of a series circuit include:

- The current is the same through every component in the loop.
- The total resistance is the sum of all individual resistances ($R_{total} = R_1 + R_2 + \dots$).
- The total source voltage is equal to the sum of the individual voltage drops.
- If the path is broken at any point, the entire circuit stops functioning.

8. Differentiate between the internal circuit of a voltmeter and an ammeter.

A voltmeter is designed with a **very high internal resistance** to ensure it does not draw significant current from the circuit when connected in parallel. Conversely, an ammeter is designed with a **very low internal resistance** (ideally zero) so that it does not significantly drop voltage when inserted in series with the load.

9. In which combination do we connect a DMM to measure current and voltages?

To measure **voltage**, the DMM must be connected in **parallel** (across the component). To measure **current**, the DMM must be connected in **series** (breaking the circuit and inserting the meter into the path).